



SAFETY DATA SHEET

ZG-S

SECTION 1. PRODUCT AND COMPANY IDENTIFICATION

Product name : ZG-S

Other means of identification : Not applicable

Recommended use : Cleaning product

Restrictions on use : Reserved for industrial and professional use.

Product dilution information : Product is sold ready to use.

Company : Ecolab Inc.
1 Ecolab Place
St. Paul, Minnesota USA 55102
1-800-352-5326

Emergency health information : 1-800-328-0026 (US/Canada), 1-651-222-5352 (outside US)

Issuing date : 08/29/2019

SECTION 2. HAZARDS IDENTIFICATION

GHS Classification

Eye irritation : Category 2B

GHS label elements

Signal Word : Caution

Hazard Statements : Causes eye irritation.

Precautionary Statements : **Prevention:**
Do not get in eyes, on skin, or on clothing. Wash skin thoroughly after handling. Do not eat, drink or smoke when using this product.
Response:
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical advice/ attention. Take off contaminated clothing and wash it before reuse. IF ON SKIN: Wash with plenty of water. If skin irritation or rash occurs: Get immediate medical advice/ attention.

Other hazards : None known.

SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

Pure substance/mixture : Mixture

Chemical name	CAS-No.	Concentration (%)
Poly(hexamethyleneguanide) hydrochloride	27083-27-8	0.089
Octyl decyl dimethyl ammonium chloride	32426-11-2	0.1333
Didecyl Dimethyl Ammonium Chloride	7173-51-5	0.0799
Diocetyl dimethyl ammonium chloride	5538-94-3	0.0534
Alkyl (50% C14, 40% C12, 10% C16) dimethyl benzyl ammonium chloride	68424-85-1	0.1778

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SECTION 4. FIRST AID MEASURES

In case of eye contact	: Rinse with plenty of water.
In case of skin contact	: Rinse with plenty of water.
If swallowed	: Rinse mouth. Get medical attention if symptoms occur.
If inhaled	: Get medical attention if symptoms occur.
Protection of first-aiders	: No special precautions are necessary for first aid responders.
Notes to physician	: Treat symptomatically.
Most important symptoms and effects, both acute and delayed	: See Section 11 for more detailed information on health effects and symptoms.

SECTION 5. FIRE-FIGHTING MEASURES

Suitable extinguishing media	: Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Unsuitable extinguishing media	: None known.
Specific hazards during fire fighting	: Not flammable or combustible.
Hazardous combustion products	: Decomposition products may include the following materials: Carbon oxides
Special protective equipment for fire-fighters	: Use personal protective equipment.
Specific extinguishing methods	: Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes.

SECTION 6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures	: Refer to protective measures listed in sections 7 and 8.
Environmental precautions	: Do not allow contact with soil, surface or ground water.
Methods and materials for containment and cleaning up	: Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

SECTION 7. HANDLING AND STORAGE

Advice on safe handling	: Wash hands thoroughly after handling. In case of mechanical
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malfunction, or if in contact with unknown dilution of product, wear full Personal Protective Equipment (PPE).

Conditions for safe storage : Keep out of reach of children. Store in suitable labeled containers.

Storage temperature : 0 °C to 50 °C

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Ingredients with workplace control parameters

Contains no substances with occupational exposure limit values.

Engineering measures : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

Personal protective equipment

Eye protection : No special protective equipment required.

Hand protection : No special protective equipment required.

Skin protection : No special protective equipment required.

Respiratory protection : No personal respiratory protective equipment normally required.

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	: absorbed liquid (wipes, pads, ...)
Color	: white
Odor	: mild
pH	: 10.5 - 12.5, (100 %)
Flash point	: Not applicable, Does not sustain combustion.
Odor Threshold	: No data available
Melting point/freezing point	: No data available
Initial boiling point and boiling range	: No data available
Evaporation rate	: No data available
Flammability (solid, gas)	: No data available
Upper explosion limit	: No data available
Lower explosion limit	: No data available
Vapor pressure	: No data available
Relative vapor density	: No data available
Relative density	: No data available
Water solubility	: No data available
Solubility in other solvents	: No data available

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Partition coefficient: n-octanol/water	: No data available
Autoignition temperature	: No data available
Thermal decomposition	: No data available
Viscosity, kinematic	: No data available
Explosive properties	: No data available
Oxidizing properties	: No data available
Molecular weight	: No data available
VOC	: No data available

SECTION 10. STABILITY AND REACTIVITY

Reactivity	: No dangerous reaction known under conditions of normal use.
Chemical stability	: Stable under normal conditions.
Possibility of hazardous reactions	: No dangerous reaction known under conditions of normal use.
Conditions to avoid	: None known.
Incompatible materials	: None known.
Hazardous decomposition products	: In case of fire hazardous decomposition products may be produced such as: Carbon oxides

SECTION 11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

Potential Health Effects

Eyes	: Causes eye irritation.
Skin	: Health injuries are not known or expected under normal use.
Ingestion	: Health injuries are not known or expected under normal use.
Inhalation	: Health injuries are not known or expected under normal use.
Chronic Exposure	: Health injuries are not known or expected under normal use.

Experience with human exposure

Eye contact	: Redness, Irritation
Skin contact	: No symptoms known or expected.
Ingestion	: No symptoms known or expected.
Inhalation	: No symptoms known or expected.

Toxicity

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Product

Acute oral toxicity	: No data available
Acute inhalation toxicity	: No data available
Acute dermal toxicity	: No data available
Skin corrosion/irritation	: No data available
Serious eye damage/eye irritation	: Mild eye irritant
Respiratory or skin sensitization	: No data available
Carcinogenicity	: No data available
Reproductive effects	: No data available
Germ cell mutagenicity	: No data available
Teratogenicity	: No data available
STOT-single exposure	: No data available
STOT-repeated exposure	: No data available
Aspiration toxicity	: No data available

Components

Acute oral toxicity	: Octyl decyl dimethyl ammonium chloride LD50 Rat: 238 mg/kg
	Didecyl Dimethyl Ammonium Chloride LD50 Rat: 329 mg/kg
	Diocetyl dimethyl ammonium chloride LD50 Rat: 238 mg/kg
	Alkyl (50% C14, 40% C12, 10% C16) dimethyl benzyl ammonium chloride LD50 Rat: 344 mg/kg

Components

Acute inhalation toxicity	: Poly(hexamethylenebiguanide) hydrochloride LC50 Rat: 0.37 mg/l Test atmosphere: dust/mist
	Octyl decyl dimethyl ammonium chloride 4 h LC50 Rat: 0.07 mg/l Test atmosphere: dust/mist
	Didecyl Dimethyl Ammonium Chloride 4 h LC50 Rat: 0.07 mg/l Test atmosphere: dust/mist
	Diocetyl dimethyl ammonium chloride 4 h LD50 Rat: 0.07 mg/l Test atmosphere: dust/mist

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Alkyl (50% C14, 40% C12, 10% C16) dimethyl benzyl ammonium chloride
4 h LC50 Rat: 0.054 mg/l
Test atmosphere: dust/mist

Components

Acute dermal toxicity : Poly(hexamethylenebiguanide) hydrochloride
LD50 Rat: > 5,000 mg/kg

Octyl decyl dimethyl ammonium chloride
LD50 Rabbit: 2,930 mg/kg

Didecyl Dimethyl Ammonium Chloride
LD50 Rabbit: 2,930 mg/kg

Diocetyl dimethyl ammonium chloride
LD50 Rabbit: 259 mg/kg

Alkyl (50% C14, 40% C12, 10% C16) dimethyl benzyl ammonium chloride
LD50 Rabbit: 3,340 mg/kg

SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity

Environmental Effects : Toxic to aquatic life with long lasting effects.

Product

Toxicity to fish : No data available

Toxicity to daphnia and other aquatic invertebrates : No data available

Toxicity to algae : No data available

Components

Toxicity to fish : Octyl decyl dimethyl ammonium chloride
96 h LC50 Fish: 1 mg/l

Didecyl Dimethyl Ammonium Chloride
96 h LC50 Fish: 1 mg/l

Diocetyl dimethyl ammonium chloride
96 h LC50 Oncorhynchus mykiss (rainbow trout): 0.35 mg/l

Components

Toxicity to daphnia and other aquatic invertebrates : Diocetyl dimethyl ammonium chloride
96 h LC50: 0.073 mg/l

Alkyl (50% C14, 40% C12, 10% C16) dimethyl benzyl ammonium chloride
48 h EC50 Daphnia magna (Water flea): 0.016 mg/l

Components

Toxicity to algae : Diocetyl dimethyl ammonium chloride
72 h EC50 Pseudokirchneriella subcapitata (algae): 0.122 mg/l

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Persistence and degradability

Not applicable - Biocide

Bioaccumulative potential

No data available

Mobility in soil

No data available

Other adverse effects

No data available

SECTION 13. DISPOSAL CONSIDERATIONS

- Disposal methods : Do not contaminate ponds, waterways or ditches with chemical or used container. Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.
- Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

SECTION 14. TRANSPORT INFORMATION

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

Land transport (DOT)

Not dangerous goods

Sea transport (IMDG/IMO)

Not dangerous goods

SECTION 15. REGULATORY INFORMATION

EPCRA - Emergency Planning and Community Right-to-Know

CERCLA Reportable Quantity

This material does not contain any components with a CERCLA RQ.

SARA 304 Extremely Hazardous Substances Reportable Quantity

This material does not contain any components with a section 304 EHS RQ.

SARA 311/312 Hazards : Serious eye damage or eye irritation

SARA 302 : This material does not contain any components with a section 302 EHS TPQ.

SARA 313 : This material does not contain any chemical components with known CAS numbers that exceed the threshold (De Minimis) reporting levels established by SARA Title III, Section 313.

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California Prop. 65

This product does not contain any chemicals known to the State of California to cause cancer, birth, or any other reproductive defects.

California Cleaning Product Right to Know Act of 2017 (SB 258)

This regulation applies to this product.

Chemical Name	CAS-No.	Function	List(s)
water	7732-18-5	Diluent	Not Applicable
Substituted carboxylic acid	Withheld	Buffer	Not Applicable
Substituted alkylamine	Withheld	Solvent	Not Applicable
Alkyl (50% C14, 40% C12, 10% C16) dimethyl benzyl ammonium chloride	68424-85-1	Biocide	Not Applicable
Octyl decyl dimethyl ammonium chloride	32426-11-2	Biocide	Not Applicable
Nonionic surfactant	Withheld	Cleaning Agent	Not Applicable
Poly(hexamethylenebiguanide) hydrochloride	27083-27-8	Biocide	Not Applicable
Didecyl Dimethyl Ammonium Chloride	7173-51-5	Biocide	Not Applicable
Dioctyl dimethyl ammonium chloride	5538-94-3	Biocide	Not Applicable

*refer to ecolab.com/sds for electronic links to designated lists

The ingredients of this product are reported in the following inventories:

United States TSCA Inventory :

On the inventory, or in compliance with the inventory

Canadian Domestic Substances List (DSL) :

This product contains one or several components that are not on the Canadian DSL nor NDSL.

Australia Inventory of Chemical Substances (AICS) :

not determined

New Zealand. Inventory of Chemical Substances :

not determined

Japan. ENCS - Existing and New Chemical Substances Inventory :

not determined

Korea. Korean Existing Chemicals Inventory (KECI) :

not determined

Philippines Inventory of Chemicals and Chemical Substances (PICCS) :

not determined

China. Inventory of Existing Chemical Substances in China (IECSC) :

not determined

Taiwan Chemical Substance Inventory (TCSI) :

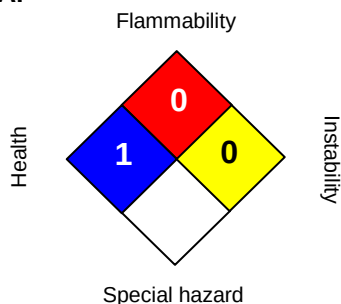
not determined

SECTION 16. OTHER INFORMATION

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NFPA:



HMIS III:

HEALTH	1
FLAMMABILITY	0
PHYSICAL HAZARD	0

0 = not significant, 1 = Slight,
2 = Moderate, 3 = High
4 = Extreme, * = Chronic

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Prepared by : Regulatory Affairs

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.